

Sai Manikanta Karnati

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Summary — Full-Stack Software Engineer with **4+ years of experience** designing and delivering **Service-Oriented Architecture (SOA)** with strong expertise in **Python, Angular, SQL development** and **RESTful web services** across enterprise and cloud-native environments. Proficient in TypeScript, Java Spring Boot, and Node.js for building scalable, modular applications. Deep expertise in **API design, versioning, and isolation** following OpenAPI/Swagger standards, with hands-on experience building **GraphQL APIs** and **microservices** deployed on **AWS**. Skilled in event-driven messaging (Kafka), and CI/CD pipelines, with a strong track record of delivering performance-optimized, production-grade systems.

Experience

Vanguard

08/2025 – present

Software Engineer

- Designed and developed scalable backend services using **Python and Java Spring Boot microservices**, enabling modular service-oriented architecture and independent deployments.
- **Built Python-based data processing pipelines** and automation scripts for backend workflows, reducing manual operational effort and improving processing efficiency.
- Developed and optimized complex SQL queries, stored procedures, and data models across relational and NoSQL databases, improving reporting and transaction performance by 35%.
- Engineered high-throughput event-driven systems using DynamoDB Streams, Python/Node.js Lambda functions, and SNS/SQS, ensuring 99.99% reliability.
- Designed AI-powered backend services integrating Python-based LLM APIs for intelligent text processing, classification, and automated workflows.
- Built responsive Angular single-page applications using TypeScript, RxJS, Angular Material, Reactive Forms, and lazy loading.
- Developed RESTful and GraphQL APIs consumed by Angular frontends, ensuring seamless frontend-backend integration.
- Optimized application performance through SQL tuning, indexing strategies, and database schema improvements.
- Deployed cloud-native applications on AWS (Lambda, ECS, EC2, S3, RDS, DynamoDB) and integrated Azure services. Implemented monitoring and observability using Splunk and Spring Boot Actuator, reducing production issue resolution time by 40%.

Xebia

01/2022 – 12/2023

Associate Software Engineer

- Developed backend applications using Python, Java Spring Boot, and Node.js, delivering scalable RESTful APIs and microservices.
- Built Python automation scripts for data transformation, validation, and backend task orchestration.
- Designed and optimized SQL queries, joins, views, and stored procedures, improving API response times by 35%.
- Built dynamic frontend applications using Angular, TypeScript, SCSS, Angular Router, RxJS, and Angular Material.
- Developed reusable Angular components and implemented state management for complex UI workflows.
- Created and documented RESTful APIs using Swagger/OpenAPI, enforcing API versioning and backward compatibility.
- Developed GraphQL APIs to improve client-side data retrieval efficiency.
- Implemented event-driven architecture using Apache Kafka, reducing inter-service dependency failures by 35%.
- Integrated authentication using OAuth2, JWT, and Spring Security.
- Used Redis caching and SQL optimization techniques to improve throughput under heavy load.
- Automated infrastructure provisioning with Terraform on AWS and GCP.
- Containerized applications using Docker and Kubernetes, improving CI/CD deployment efficiency.

Xebia

01/2020 – 12/2021

Engineer Trainee

- Developed enterprise applications using Angular and Python/Java backend services.
- Built and optimized SQL queries, stored procedures, and relational database schemas for transactional systems.
- Worked extensively with MySQL and MongoDB, improving query execution time by 20%.
- Developed RESTful APIs and documented service contracts using Swagger/OpenAPI.
- Built Angular frontend modules using TypeScript, Redux, and Reactive Forms.
- Implemented API security using OAuth2 and Spring Security.
- Applied rate limiting and API gateway protections for backend services.
- Increased unit test coverage to 95% using JUnit and Python-based test automation tools.

Education

University Of North Carolina At Charlotte

Master of Science in Computer Science

Certifications

- AWS - AWS Certified Developer Associate
- MongoDB - MongoDB Certified Developer
- Infosys - Infosys Certified Software Programmer
- Java - NPTEL certified Java Developer
- AWS - AWS Academy Graduate - AWS Academy Data Engineering

Skills

Frontend: Angular, TypeScript, HTML5, CSS3, SASS/SCSS, RxJS, Angular Material, Vue.js, React.js

Backend: Python, Java Spring Boot, Spring MVC, Spring Security, Spring Cloud, Node.js, Express.js, RESTful APIs, GraphQL, Meteor JS

API & Arch: Service-Oriented Architecture (SOA), Microservices, API Versioning, OpenAPI/Swagger, JSON, REST, GraphQL

Databases: MySQL, MongoDB, Redis, DynamoDB

Languages: JavaScript, TypeScript, Java, Python, SQL

Cloud: AWS (EC2, S3, Lambda, ECS, API Gateway), Azure, Google Cloud Platform (GCP)

DevOps: Docker, Kubernetes, Terraform, CI/CD, Jenkins

Messaging: Apache Kafka, AWS SNS/SQS

Tools: Git, BitBucket, JIRA, Postman, Splunk, Confluence

Methodologies: Agile, Scrum, TDD, SOA, Microservices Architecture